

Special Communication

Evaluating an information theoretic approach for selecting multimodal data fusion methods

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ABSTRACT

Objective: Interest has grown in combining radiology, pathology, genomic, and clinical data to improve the accuracy of diagnostic and prognostic predictions toward precision health. However, most existing works choose their datasets and modeling approaches empirically and in an ad hoc manner. A prior study proposed four partial information decomposition (PID)-based metrics to provide a theoretical understanding of multimodal data interactions: redundancy, uniqueness of each modality, and synergy. However, these metrics have only been evaluated in a limited collection of biomedical data, and the existing work does not elucidate the effect of parameter selection when calculating the PID metrics. In this work, we evaluate PID metrics on a wider range of biomedical data, including clinical, radiology, pathology, and genomic data, and propose potential improvements to the PID metrics.

Methods: We apply the PID metrics to seven different modality pairs across four distinct cohorts (datasets). We compare and interpret trends in the resulting PID metrics and downstream model performance in these multimodal cohorts. The downstream tasks being evaluated include predicting the prognosis (either overall survival or recurrence) of patients with non-small cell lung cancer, prostate cancer, and glioblastoma.

Results: We found that, while PID metrics are informative, solely relying on these metrics to decide on a fusion approach does not always yield a machine learning model with optimal performance. Of the seven different modality pairs, three had poor (0%), three had moderate (66%–89%), and only one had perfect (100%) consistency between the PID values and model performance. We propose two improvements to the PID metrics (determining the optimal parameters and uncertainty estimation) and identified areas where PID metrics could be further improved.

Conclusion: The current PID metrics are not accurate enough for estimating the multimodal data interactions and need to be improved before they can serve as a reliable tool. We propose improvements and provide suggestions for future work. Code: <https://github.com/zhtyolivia/pid-multimodal>.

1. Introduction

Multimodal data, spanning clinical, radiologic, pathologic, molecular, and other data types, are increasingly captured in electronic health records. Combining different data sources provides a more comprehensive characterization of the individual's condition. Utilizing artificial intelligence (AI) and machine learning (ML) techniques to integrate these different modalities promises to improve the accuracy of diagnostic and prognostic tasks [1]. A growing number of works has shown the utility of multimodal data fusion, including a weakly supervised learning-based multimodal deep learning framework for integrating histopathology and genomic data for cancer patient prognosis in 14

different cancer types [2], a graph neural network-based framework for integrating pathology, genomic, and clinical information to predict overall survival in six cancer types [3], and an ML pipeline for integrating radiology and pathology image features to predict therapy response in locally advanced rectal cancer [4]. One challenge of multimodal modeling is choosing the optimal approach for fusing features across modalities; the selection of approach is commonly done empirically and in an ad hoc manner. For instance, while researchers often perform feature selection, dimensionality reduction, and early-, intermediate-, or late-fusion, the choice of features and fusion methods is done in an ad hoc manner.

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Recently, Liang et al. [5] proposed using information theory to quantify multimodal interactions. Examining individual modalities in a pairwise manner, they presented four PID-based metrics to quantify the interactions that are redundant between two modalities (R), unique to the first modality ($U1$), unique to the second modality ($U2$), and the synergy when combining the modalities (S). The authors evaluated the PID metrics on synthetic data, pathology-genomics fusion data, mental health, and robotics data and demonstrated using the relative values of the PID metrics as a guide to choose the appropriate data and modeling types. If $U1$ or $U2$ is the highest, then training a model corresponding to that single modality is preferred; if R is the highest, a late fusion-based approach combining the two modalities is suggested; if S is the highest, an early-fusion-based approach for combining the two modalities is recommended.

Several limitations of the work in Liang et al. [5] are noted. First, the evaluation of PID metrics in the biomedical domain was limited to pathology-genomics and mental health datasets. A wider spectrum of multimodal combinations, such as integrating radiological images and clinical data, was not analyzed in their work. Various established clinical models use clinical factors as input and predict the aggressiveness of various diseases such as prostate cancer [6] and breast cancer [7], underscoring the value of evaluating PID metrics on clinical and radiology features as well. Second, Liang et al. [5]’s results on the two biomedical datasets were presented as case studies where they briefly commented on the trends of the PID values and model performance but did not formally quantify the degree of consistency between the PID metrics and model performance. In our study, we not only present the trends but also provide the percentage of PID values that are consistent with model performance in each of the modality pair. Third, they only explored a limited number of data fusion techniques (tensor fusion) and modeling choices (deep learning model). Our study addresses that limitation by (1) incorporating two more commonly used early-fusion techniques and (2) using non-deep-learning and deep-learning-based prediction models for each modality. Doing so can help us get a more comprehensive list of model results to compare against the PID metrics and gain more insights into how well the PID metrics can capture the multimodal data interactions. Fourth, Liang et al. [5] did not mention anything regarding the effect of preprocessing parameters on the PID values. We sought to address that by experimenting with the two important parameters when computing the PID values and proposing suggestions on the choices of these parameters. Last but not least, PID values were not calculated with any confidence intervals associated with them, making it difficult to assess the potential variability in the PID values when determining which value is greatest. We sought to address that by performing bootstrapping to derive a 95% confidence interval (CI) for each PID value.

In this work, we apply the PID metrics to seven different modality pairs across four distinct multimodal radiology cohorts (datasets). We compare and interpret trends in the resulting PID metrics and downstream model performance in these multimodal radiology cohorts. The downstream tasks being evaluated include predicting the prognosis (either overall survival or recurrence) of patients with non-small cell lung cancer, prostate cancer, and glioblastoma. We hypothesize that the four PID metrics are consistent with the downstream modeling performance. Fig. 1 visualizes the overview of this project.

Our contributions are as follows:

1. As compared to Liang et al. [5], we evaluate PID metrics on a wider range of biomedical data, including radiology-pathology, radiology-clinical, and radiology-genomic data. We also expanded the types of models to compare the PID metrics with, allowing a more comprehensive investigation of the consistency between the PID metrics and various modeling techniques.

2. Besides reporting the general trends like Liang et al. [5] did, we also quantified the degree of consistency between the PID values and model performance.

3. We propose suggestions for choosing the appropriate preprocessing parameters when computing the PID values and propose means to estimate the uncertainty associated with the PID values.

The organization of this work is as follows: we will first present background information on multimodal interaction measures and formally describe the PID metrics. We will then describe each dataset, modeling techniques, and experiment setups. We show results and discuss their implications in Table 1, “Statement of significance”.

2. Methods

2.1. Existing works on interaction measures

Prior to PID metrics, non-information-theory approaches have been used to capture the interaction among features, such as Shapley values, integrated gradients (IG), and canonical correlation analysis (CCA). However, these three approaches cannot generate the estimation for all four interactions (redundancy, uniqueness of modality 1, uniqueness of modality 2, and synergy). Briefly, Shapley values are based on interactions captured by a multimodal model [8]. They only capture the overall independence of each modality instead of redundancy, uniqueness, or synergy. It is also usually difficult to estimate Shapley values for large, high-dimensional datasets efficiently [5]. IG is limited to capturing only the unimodal contribution of each modality in a trained model [9] by examining the importance score of each modality. However, PID enables one to estimate the uniqueness of each modality before any model training. CCA [10] can be applied to establish the correlation between two modalities, but they do not measure the uniqueness of each modality. Overall, the PID estimate is more advantageous in that it can quantify all four interactions in a computationally efficient manner and can do so without any downstream model training, serving as prior guidance on how to best utilize data from different modalities in the subsequent model training.

2.2. PID estimation

Traditionally, information theory, particularly the measurement of mutual information, provides a framework for reasoning about the information obtained about one variable by observing another. However, the generalization of mutual information to multiple variables, also known as interaction information, can have negative values, which are counterintuitive to interpret. Motivated by this observation, Williams et al. [11] proposed the theory of PID, which decomposes the contributions of multiple variables into positive and interpretable terms. Liang et al. [5] utilized these decomposed terms in multimodal data fusion by treating them as metrics for analyzing the contribution of each modality and introduced methods for efficient computation of these metrics. In particular, as shown in Fig. 2, given two sets of features representing two different modalities, $X1$ and $X2$, PID decomposes their total information about the outcome Y , $I(X1, X2; Y)$ into: redundancy R , uniqueness $U1$ or $U2$, and synergy S . Redundancy R is the intersection between the information provided by $X1$ and $X2$; uniqueness $U1$ or $U2$ is information contributed by only $X1$ or $X2$; synergy S denotes information that only arises when considering both $X1$ and $X2$. In the context of multimodal learning, such decomposition is a powerful tool for reasoning about the roles of various modalities in the final prediction. Let p be the joint distribution over $X1$, $X2$, and Y , the decomposition terms are defined as:

$$R = \max_q I_q(X1; X2; Y) \quad (1)$$

$$U1 = \min_q I_q(X1; Y|X2) \quad (2)$$

$$U2 = \min_q I_q(X2; Y|X1) \quad (3)$$

$$S = I_p(X1, X2; Y) - \min_q I_q(X1, X2; Y) \quad (4)$$

Table 1
Statement of significance.

Problem or Issue	Objective measures are needed to guide the selection of multimodal data fusion approaches.
What is Already Known	Partial information decomposition (PID)-based metrics can provide theoretical guidance on how to perform multimodal data fusion: redundancy and synergy between two modalities and uniqueness of each modality.
What this Paper Adds	We evaluate the PID metrics to develop multimodal models using combinations of clinical, radiology, pathology, and genomic data. While informative, following the guidance of PID metrics did not always yield the optimal model. We quantify the degree of consistency between the PID values and model performance. We explore two improvements to the PID metric (determining the optimal parameters and uncertainty estimation) and identify areas for further improvement.
Who would benefit from the new knowledge in this paper	Researchers developing multimodal AI models require objective measures for assessing types of fusion methods.

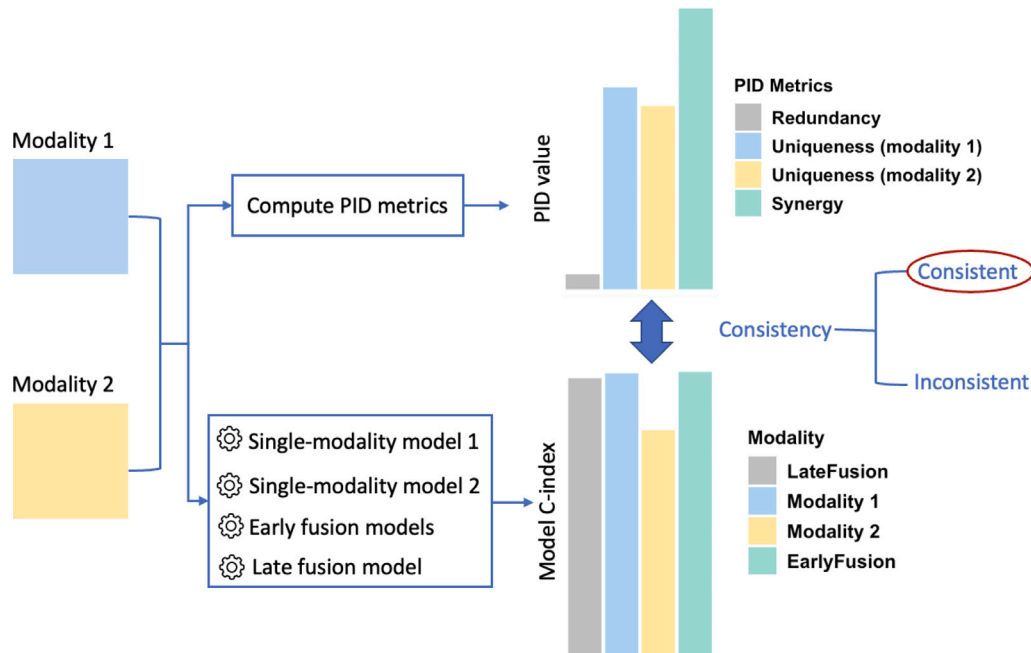


Fig. 1. The overview of this study. The two modalities are passed through two main modules, a PID metrics computation module, and a modeling module, to obtain the four PID metrics' values and the model performance, as measured by C-indices. Consistency between the four metrics and C-indices is then examined. The shared color between the bars of the PID metrics and C-indices indicates an expected consistency between them. For example, synergy and EarlyFusion have the same bar color since one expects that if synergy is the highest among the PID metrics, then the C-index of the early fusion model should also be the highest among all models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

which follows definitions in Bertschinger et al. [12] with q being in the set of distributions on X_1 , X_2 , and Y satisfying the marginal constraint $q(x_i, y) = p(x_i, y) \forall x \in X_i, y \in Y$, and $i \in \{1, 2\}$.

Liang et al. [5] proposed two scalable solvers for addressing the optimization problem in Eqs. (1) to (4). In particular, when the variables are discrete with a small, finite set of possible values (e.g. around 100), exact solutions can be obtained via a convex solver called CVX. With high-dimensional and continuous variables, the authors formulated an approximator based on neural networks. We used CVX in all the experiments in our study. Solving Eqs. (1) to (4) is equivalent to solving a max-entropy optimization problem $q = \arg \max_{q \in \Delta p} H_q(Y|X_1, X_2)$ where $H_q(Y|X_1, X_2)$ represents the conditional entropy of Y given X_1 and X_2 , which measures the uncertainty or entropy of Y when the values of X_1 and X_2 are known. Directly solving for q is numerically difficult: Liang et al. [5] overcame this issue by solving a Kullback–Liebler-divergence problem $q^*, \tilde{q}^* = \arg \max_{q, \tilde{q}} \log |Y| - KL(q||\tilde{q})$ with $\tilde{q}$ being an auxiliary density function satisfying $\tilde{q}(x_1, x_2, y) = q(x_1, x_2)/$

$|Y|$. Other marginal constraints are also enforced via linear constraints: $\sum_{x_2} q = p(x_1, y)$, $\sum_{x_1} q = p(x_2, y)$, and $\sum_{x_1, x_2, y} q = 1$. Such formulation can be solved using common convex solvers. The solution q^* can be plugged back into Eqs. (1) to (4) to obtain the exact values of the decomposition.

During PID calculation, the CVX solver requires the input modalities and prediction labels to be discretized. In their work, [5] used PCA first to reduce the dimensionality of the original modalities and then used the k-means clustering (k-means) method on the dimensionality-reduced data to obtain the cluster indices of each patient, which were used as one of the inputs to the CVX solver. As for the prediction labels, [5] discretized the survival time of patients via histogramming and obtained the bin labels of each patient as input to the CVX solver (see Section 2.6 for more details). Fig. 3 shows the workflow of computing the PID metrics.

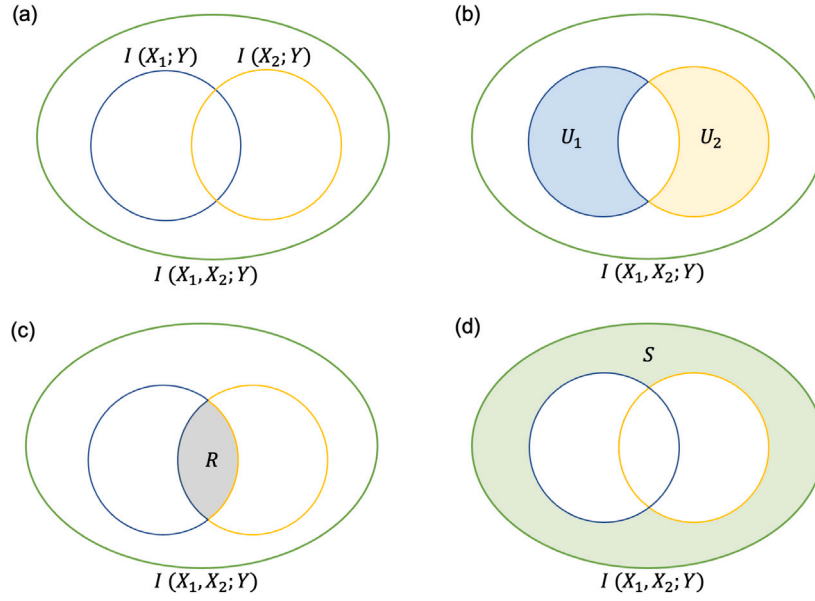


Fig. 2. Visualization of the four PID metrics (U_1 , U_2 , R , and S). In figure (a), the inner circles represent the information that modalities X_1 and X_2 provide about the task Y , respectively. The outer circle represents $I(X_1, X_2; Y)$, the total information that X_1 and X_2 together provide about Y . (b) illustrates U_1 (blue) and U_2 (yellow), the unique information that X_1 and X_2 independently provide about the task Y . (c) shows R (gray), the shared information in X_1 and X_2 about Y . (d) represents S (green), the synergy that arises only when X_1 and X_2 are combined to predict Y . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

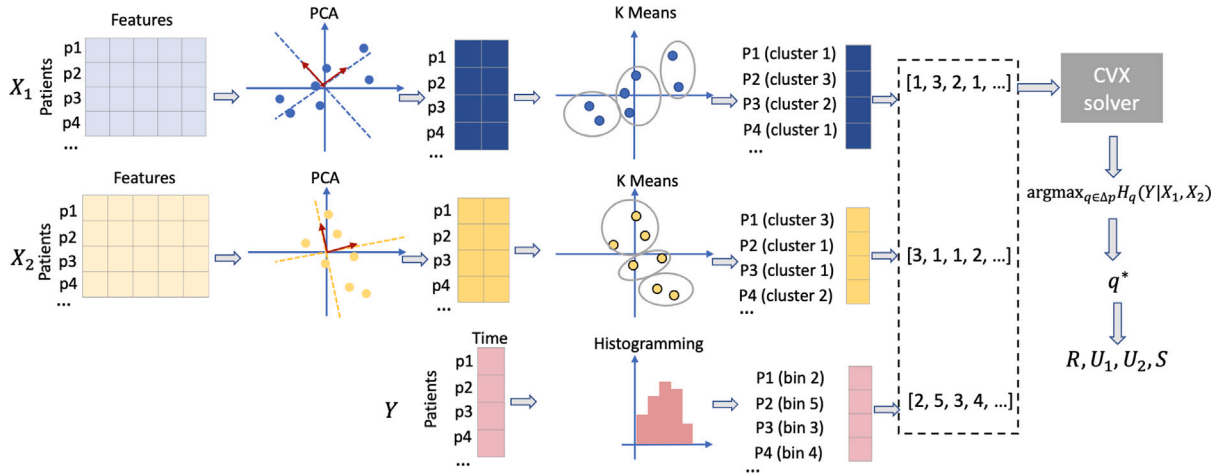


Fig. 3. Workflow of the computation of the four PID metrics.

2.3. Datasets description

We used four multimodal datasets in this study: (1) lung cancer radiomic-pathomic data predicting recurrence in early-stage lung adenocarcinoma (LUAD), denoted as *LungRadiomicPathomic*, (2) lung cancer radiogenomic data predicting overall survival in non-small cell lung cancer (NSCLC), denoted as *LungRadiogenomic*, (3) prostate cancer T2W-ADC-clinical data predicting recurrence, denoted as *ProstateT2W-ADC*, *ProstateCAPRAS-T2W*, *ProstateCAPRAS-ADC*, and *ProstateCAPRAS-Imaging*, and (4) glioblastoma (GBM) radiogenomic data for predicting overall survival, denoted as *BrainRadiogenomic*. These datasets were used because they cover different diseases, commonly used modalities in the medical domain (radiomic, pathomic, genomic, and clinical data), and different modality pairs and are publicly available except for the prostate data.

2.3.1. LungRadiomicPathomic ($N = 106$)

The patient-level matched pre-surgical computed tomography (CT) scans and whole slide images (WSIs) were obtained from the National

Lung Screening Trial (NLST) [13]. We focused on a cohort of 106 patients with early-stage LUAD who underwent surgical treatment. Among these patients, 26 (24.5%) experienced disease progression (i.e., LUAD recurrence or lung cancer-related deaths). Each patient had 456 radiomic features consisting of 256 deep features extracted using a pretrained lung nodule malignancy prediction model [14] and 200 traditional features extracted using PyRadiomics [15]. The traditional radiomic features included 107 shape, intensity, and texture features from the tumoral region and 93 intensity and texture features from the peritumoral region. CT scans were normalized using a generative adversarial network before feature extraction [16]. From the WSIs, tile-level pathomic features included 512 deep features extracted from the last fully connected layer of a pretrained ResNet18 classifier for LUAD histologic subtypes [17] and 27 hand-crafted features [18,19]. The tile-level features were aggregated into patient-level features by taking the mean, standard deviation, and range of each feature, resulting in 1,617 pathomic features.

2.3.2. LungRadiogenomic ($N = 310$)

The lung cancer radiogenomic data was obtained from a study by Grossmann et al. [20]. The study provided radiomic and genomic features from a development cohort of North American NSCLC patients ($N = 224$) and a test cohort of European NSCLC patients ($N = 87$). Patients in both cohorts had confirmed primary tumors and underwent surgical treatment. We excluded one patient from the development cohort with a zero survival time and mixed the development and test cohorts for our analysis, resulting in $N = 310$ patients with 136 (43.9%) deaths. There are 636 radiomic features for each patient, consisting of 14 shape, 15 intensity, 44 texture, and 384 wavelet features. For genomic features, the expression of 21,766 unique genes was measured.

2.3.3. ProstateT2W-ADC, ProstateCAPRAS-T2W, ProstateCAPRAS-ADC, ProstateCAPRAS-Imaging ($N = 358$)

Among patients with clinically significant (Gleason score 3 + 4 or above) prostate cancer whose data were collected at our institution, 358 had patient-level matched pre-surgery T2-weighted (T2 W), apparent diffusion coefficient (ADC) maps from the magnetic resonance imaging (MRI), clinical information, and tumor contour annotations (from which the radiomic features were extracted) available. Biochemical recurrence (BCR) is defined by prostate-specific antigen (PSA) level greater than 0.2 ng/mL anytime after radical prostatectomy. Among the 358 patients, 84 (23.5%) had BCR. From T2 W and ADC images, 107 features (14 shape, 18 intensity, 75 texture features) were extracted using PyRadiomics [15]. In addition, six clinical features from an established clinical risk model for BCR prediction called the Cancer of the Prostate Risk Assessment (CAPRAS) [6] were used as another modality, including the PSA level, surgical margin, seminal vesicle invasion, Gleason score, extracapsular extension, and lymph node invasion.

2.3.4. BrainRadiogenomic ($N = 140$)

From the Cancer Imaging Archive and the Cancer Genome Atlas, 175 GBM patients had both transcriptomes where 12,042 genes were sequenced and pre-surgery MRIs available, out of which 140 had survival information and annotation of six MRI traits by a board-certified neuroradiologist. The six traits are categorical variables that describe the qualitative visual and phenotypic variations of the tumor on the MRI, and they include assessment of tumor enhancement, non-contrast enhancement, edema, necrosis, and abnormality size [21]. These categorical variables were one-hot encoded and resulted in 20 binary variables, which were used in the analysis. A total of 125 (89.3%) patients died, and the rest were alive or censored.

2.4. Survival analysis

Two types of downstream prediction models were used in this study, namely the linear Cox proportional hazards model with L2 (ridge) regularization [22] and the multilayer perceptron (MLP) model [23] with DeepSurv loss [24]. In the linear Cox model, features were selected by the Minimum Redundancy Maximum Relevance (mRMR) algorithm [25], which selects the features that are the most relevant to the target variable (recurrence or survival status) and are the least redundant among the selected features. In MLP, dropout, and L1 weight regularization were used to reduce overfitting.

2.5. Data fusion

As shown in Fig. 4, three commonly-used early-fusion techniques, including vector concatenation, sparse canonical correlation analysis (SCCA) [26], and tensor/Kronecker product [27], and one late fusion technique (model ensemble) were used. For vector concatenation (Fig. 4a), two modeling approaches, namely the linear Cox model and the MLP model, were respectively applied. For vector concatenation with linear Cox, features were first selected individually with mRMR from each modality, and then the selected features were concatenated. The

concatenated features were then used as model input. For vector concatenation with MLP, features from each modality were first fed into an MLP layer or directly used as the input to the concatenation operation, and the model with the best performance was kept. When applying SCCA for feature fusion (Fig. 4b), SCCA identifies a subset of variables from each modality that is maximally correlated and calculates a linear combination of these variables. For *LungRadiomicPathomic*, *LungRadiogenomic*, and *BrainRadiogenomic* datasets, due to the high dimensionality of features from these datasets, pre-selection of features was performed on each modality respectively via mRMR before SCCA. In addition, for *LungRadiogenomic* and *BrainRadiogenomic*, principal component analysis (PCA) was used to reduce the dimensionality of the genomic data to have roughly the same dimension as the radiomic modality before the mRMR pre-selection of features. For early fusion via the tensor product (Fig. 4c), features from each modality were first fed into an MLP layer, and the outputs from the MLP layers were used for the tensor product. For the late-fusion (ensemble) model (Fig. 4d), in both linear Cox and MLP versions, the predicted risk scores from the individual modalities were averaged together using a fixed weight on each modality. Given the small dimensionality of certain datasets (CAPRAS and the radiology traits in *BrainRadiogenomic*), we did not build single-modality MLP models for these modalities to avoid underfitting. Similarly, for late fusion, only the linear Cox model was used in the CAPRAS and *BrainRadiogenomic* datasets due to the absence of MLP models for one of the modalities.

2.6. Experimental setup

Ten repetitions of five-fold cross-validation were used. The hyperparameters of the linear Cox model and the MLP model, SCCA, mRMR feature selection, and individual modality weight during the ensemble were determined based on the average validation concordance index (C-index). The final performance of all models was compared using the average test-set C-index, and the relative values of the PID metrics were compared with the model performance for their consistency. Specifically, if redundancy is the highest among the PID metrics, then the C-index of the late fusion model should also be the highest among all models; if uniqueness of modality 1 is the highest among the PID metrics, then the C-index of the single-modality model of modality 1 should also be the highest among all models; if uniqueness of modality 2 is the highest among the PID metrics, then the C-index of the single-modality model of modality 2 should also be the highest among all models; if synergy is the highest among the PID metrics, then the C-index of the early fusion model should also be the highest among all models. For simplicity, across different early-fusion techniques, the one with the highest C-index was used to compare with the PID metrics. Similarly, across linear Cox and MLP versions for all models, the version with the best C-index was used for comparison with the PID metrics.

During PID calculation, the CVX solver requires the input modalities and prediction labels to be discretized. In their work, [5] used PCA first to reduce the dimensionality of the original modalities and then used the k-means method on the dimensionality-reduced data. However, the number of principal components and the number of clusters were not stated. We computed PID metrics using 3, 5, and 10 clusters. As for the number of principal components to use for each modality, by default, we used 2 and 5 components for all modalities. In addition, we computed the number of components that gave us at least 80% cumulative explained variance ratio. If that number was equal to or less than 5, we kept 2 and 5 components. If that number was greater than 5, then we used that number in addition to 2 and 5. To discretize the prediction labels, following [5], we binned the survival time of patients by setting the number of bins to be the cubed root of the number of patients.

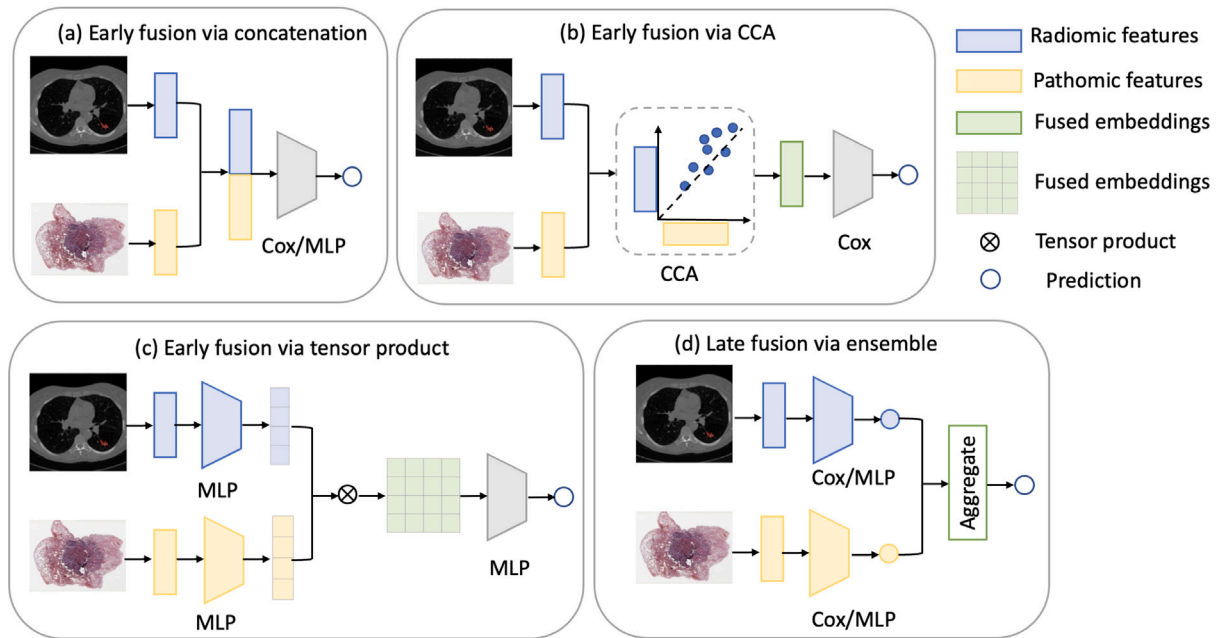


Fig. 4. Visualization of different fusion techniques used in this study.

3. Results

Fig. 5 shows the trends from both the PID values and model C-indices. As there are multiple groups of PID values from different combinations of PCA and K-means hyperparameters, for modality pairs that have a mixture of both consistent and inconsistent PID values, only one example consistent group and one example inconsistent group was respectively shown in Fig. 5 for simplicity. For modality pairs where all PID values were consistent or inconsistent, only one of such groups of values was shown. The complete results from all different PID groups across the 7 modality pairs were listed in supplementary materials Tables 1, 3, 5, 7, 9, 11, and 13, and complete results from modeling were listed in supplementary materials Tables 2, 4, 6, 8, 10, 12, and 14. Fig. 6 shows the percentage of consistent PID values out of all the parameter combinations in each of the 7 modality pairs. For example, *LungRadiomicPathomic* had 88.9% consistent PIDs since 8 out of 9 (88.9%) different groups of PIDs were consistent with the model C-indices.

Overall, the consistency between the reported model performance and PID values was mixed, depending on the pairs of modalities being examined and the task. Specifically, according to Fig. 6, only one modality pair (*ProstateCAPRAS-Imaging*) had 100% consistent PIDs; three modality pairs had moderately consistent PIDs, with *LungRadiomicPathomic* and *BrainRadiogenomic* being 88.9% and *ProstateT2W-ADC* being 66.7%; three modality pairs (*LungRadiogenomic*, *ProstateCAPRAS-T2W*, and *ProstateCAPRAS-ADC*) had zero (0%) consistency between the PID values and model performance.

The model performance of *ProstateCAPRAS-Imaging* (Fig. 5f) was perfectly consistent with all the PID values across different PID-related parameters. Specifically, across experiments, this modality pair consistently had the highest synergy compared to the other three metrics, and the C-index of the early-fusion model was also the highest, indicating that this modality pair benefited from modality fusion. In addition, in this modality pair, the uniqueness of unimodal CAPRAS data was higher than that of the imaging modality, which is consistent with the C-indices of the unimodal models.

For *BrainRadiogenomic*, the PID values were only perfectly consistent with the C-indices of the models when using the number of principal components that resulted in 80% cumulative explained variance ratio for each modality (See Supplementary Table 13 for their PID values).

This highlights the importance of choosing the optimal number of principal components so that the information loss can be minimized as much as possible during the preprocessing of PID estimation, especially for high-dimensional data like genomic data. Fig. 7 illustrates that the two genomic datasets used in this study both needed a relatively large number of principal components to meet the threshold of having an 80% explained variance ratio.

As for *LungRadiomicPathomic* and *ProstateT2W-ADC*, as shown in Fig. 6, 88.9% and 66.7% of the PID values from different PID parameter combinations were consistent with the model performance respectively in *LungRadiomicPathomic* and *ProstateT2W-ADC*. For *LungRadiomicPathomic*, all synergy values were consistent with the C-index of the early-fusion model except for 2 groups of PIDs (Supplementary materials Table 1), and a minority (2 out of 9) of the unimodal PID values were not consistent with the unimodal C-indices. For *ProstateT2W-ADC*, all synergy values were consistent with the C-index of the CCA-fused T2W-ADC model except for when the number of k-means clusters was 3 for PID, but most (4 out of 6) of uniqueness PID values were not consistent with the unimodal C-indices.

LungRadiogenomic, *ProstateCAPRAS-T2W*, and *ProstateCAPRAS-ADC* had poor consistency between the PID values and model performance (all had 0% according to Fig. 6). For *LungRadiogenomic*, redundancy was the largest compared to the other three PID metrics, indicating that late-fusion (e.g. ensemble) should give the best model performance, but that was not the case. Model performance shown in Supplementary Table 4 suggests that the early-fusion model had the best performance. For *ProstateCAPRAS-T2W*, the synergy values are all higher than the other three metrics across different parameters, indicating that the early-fusion of CAPRAS and T2 W model should have the best performance. However, the CAPRAS unimodal model had the best C-index. As for *ProstateCAPRAS-ADC*, when the number of k-means clusters was 3, the absolute value of uniqueness of CAPRAS was similar to the absolute value of synergy and the 95% CIs overlap considerably between the uniqueness of CAPRAS and synergy, indicating that either CAPRAS alone or the fused CAPRAS and T2 W features are both beneficial for the modeling task of predicting BCR. Although synergy was not consistent with the early-fusion results, CAPRAS always had a higher uniqueness than one of T2 W and ADC, which was consistent with the fact that the unimodal CAPRAS model had a better C-index than one of T2 W and ADC.

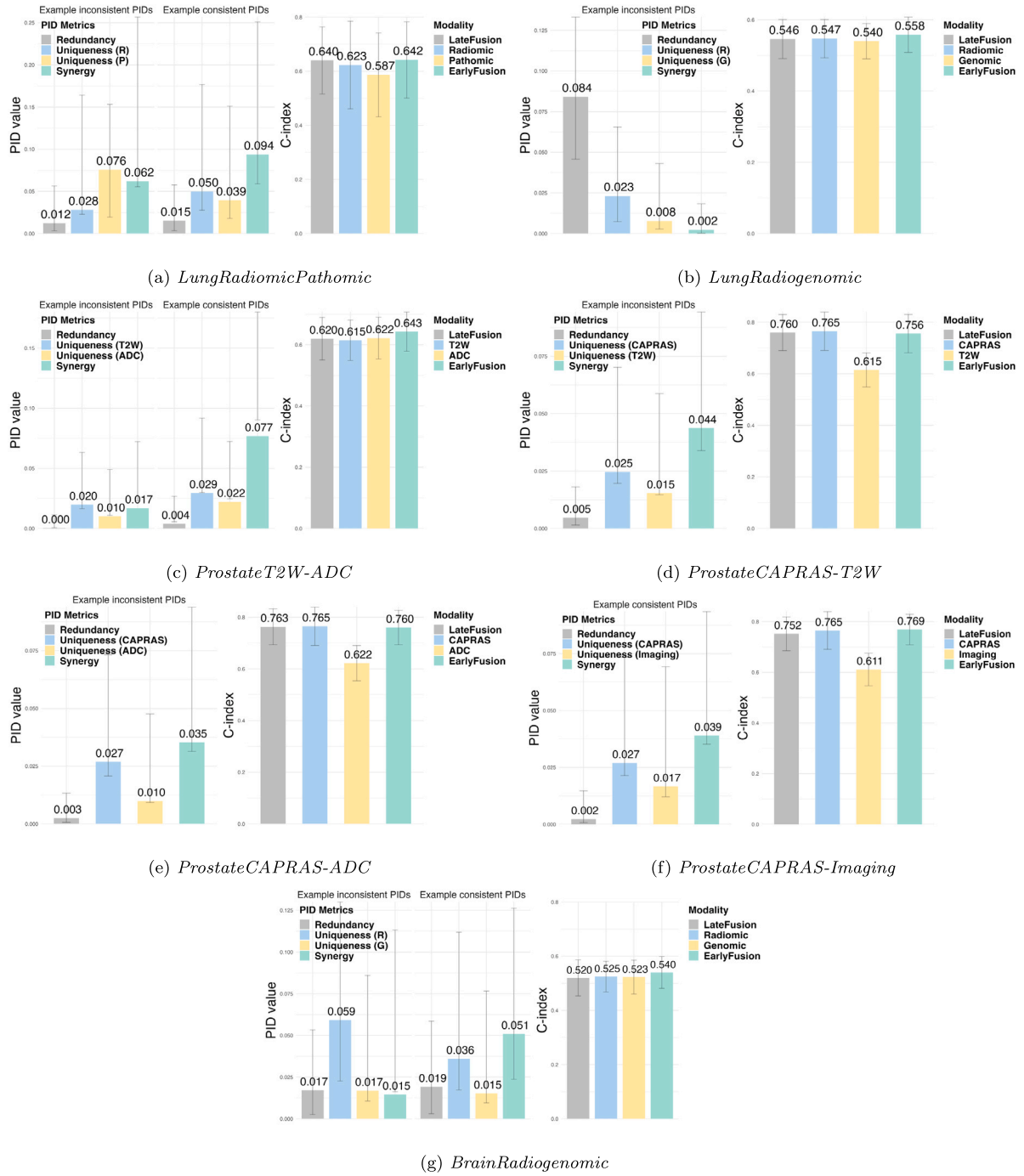


Fig. 5. Bar plots showing example PID values and the downstream modeling results represented by the average test-set C-index and its standard deviation. The error bars of the PID value bar plots indicate the 95% bootstrap CI, with the lower bound representing the 2.5% percentile and the upper bound representing the 97.5% percentile. The error bars of the C-index bar plots indicate the standard deviation of the C-indices across repeated cross-validation, with the lower bound representing mean minus standard deviation and the upper bound representing mean plus standard deviation. Figure b and d only have examples of inconsistent PIDs since none of the PIDs is consistent with the reported C-indices. Similarly, figure f only has example consistent PIDs since no group of PIDs is inconsistent with the reported C-indices.

4. Discussion

Objective measures are needed to guide the selection of multimodal data fusion approaches. Four PID metrics (redundancy, uniqueness of each modality, and synergy) have been developed, providing theoretical guidance on how to perform multimodal data fusion. According to Liang et al. [5], the PID metrics can help determine (1) whether available data from different modalities should be combined and (2) the optimal approach to combine them, which would advance the development of multimodal models and introduce new rigor in how

these models are developed. However, their original work fell short in several aspects, including the lack of quantification of the consistency between the PID values and model performance, the lack of evaluation on a diverse range of biomedical multimodal data, missing experiments on the effect of preprocessing parameters on the PID values, and missing uncertainty estimate on the PID values. We sought to address those limitations using various multimodal radiology data. In applying PID metrics to our datasets, only one out of the seven modality combinations showed 100% consistency between the PID values and

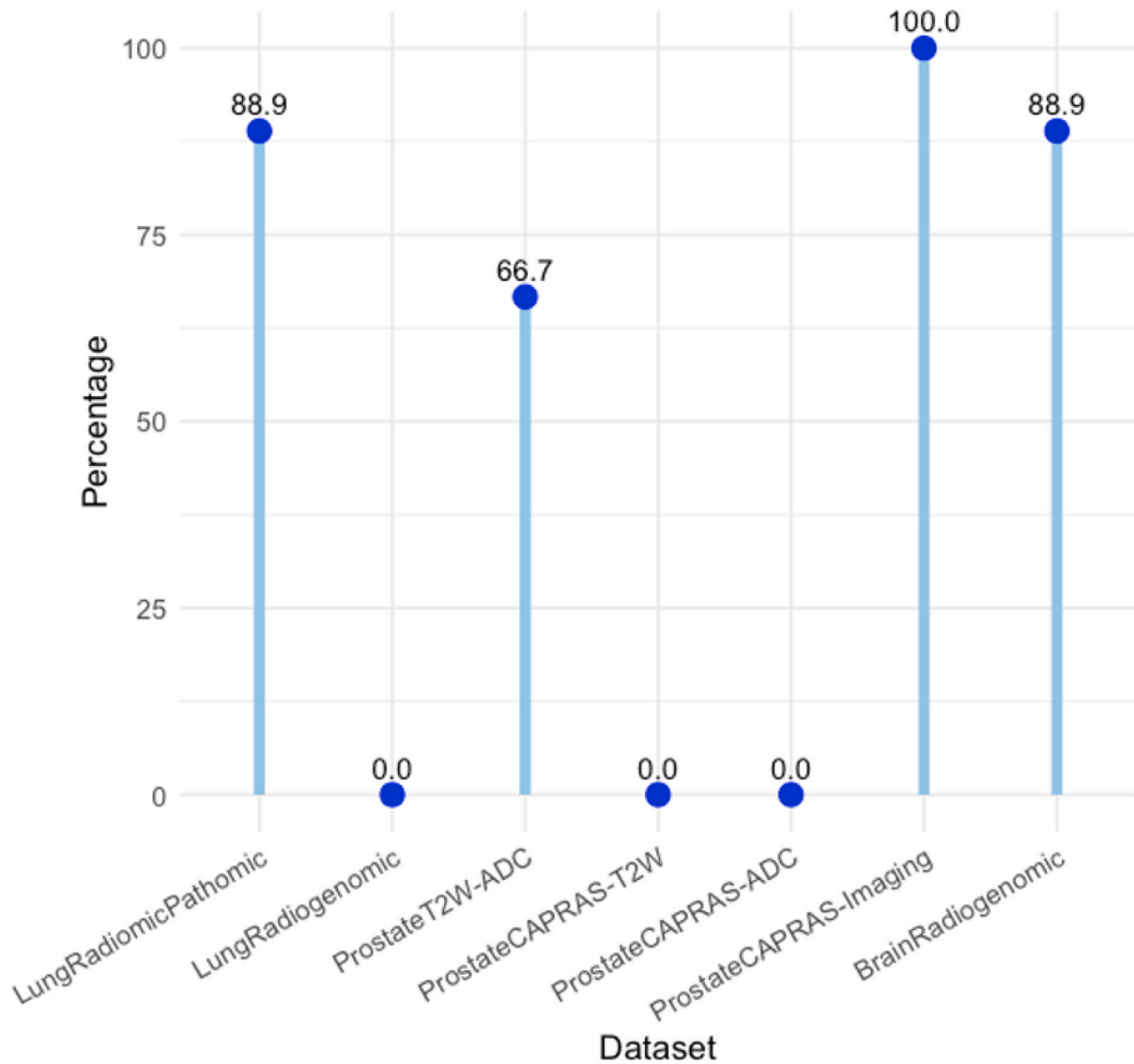


Fig. 6. Lollipop plot visualizing the percentage of parameter combinations resulting in PID values that are consistent with model performance in each of the seven modality pairs.

model performance, three modality pairs resulted in 60%–90% consistency, and three showed zero consistency. These results indicate that the current PID metrics are not accurate enough for estimating the multimodal data interactions and need to be improved before they can serve as a reliable tool.

We provide several conjectures to explain the inconsistency between the PID metrics and model performance. First, when the difference in PID metrics is small, selecting the highest metric as an indicator of optimal modeling choice can be misleading, as the metric difference may not be significantly large enough to ensure model performance. For example, in *ProstateT2W-ADC*, although the uniqueness of T2 W (0.020) is marginally higher than the synergy between T2 W and ADC (0.017), the 95% CI's null hypothesis testing indicates that the difference between them was not statistically significant. In other words, the uniqueness of T2 W is roughly the same as the synergy between T2 W and ADC, indicating that the amount of information in T2 W alone is similar to that in the combined T2 W and ADC. This means that one can either use the T2 W single modality model or the early fusion model. Since the early fusion model had the best performance, one can argue that the PID metrics are consistent with model performance. Similar trends can also be observed in *ProstateCAPRAS-T2W* and *ProstateCAPRAS-ADC*, where the uniqueness of CAPRAS is roughly

the same as the synergy between the two modalities. These observations indicate the importance of having an uncertainty measure for each PID value, as it can help one discern the differences in each value and better inform the modeling choice. Another possible reason for the inconsistency between the PID metrics and model performance could be the suboptimal compression of genomic data in the preprocessing step of PID calculation, especially in *LungRadiogenomic*, which had twice the number of genomic features as in *BrainRadiogenomic*. Although we tried to retain as much information as possible by taking into account the PCA cumulative explained variance ratio, since there are more than 20,000 genomic features in *LungRadiogenomic*, PCA might have failed to fully capture the information in the genomic features. This indicates the need to improve the preprocessing step of PID calculation in the future.

In summary, we sought to evaluate an existing PID-based metric meant to serve as theoretical guidance on multimodal data fusion approaches. Compared to Liang et al. [5], we examined PID metrics on a wider range of biomedical data, including radiology-pathology, radiology-clinical, and radiology-genomic data. We broadened the range of models used to compare PID metrics, enabling a more thorough examination of the consistency between PID metrics and different modeling approaches. Beyond reporting the general trends like

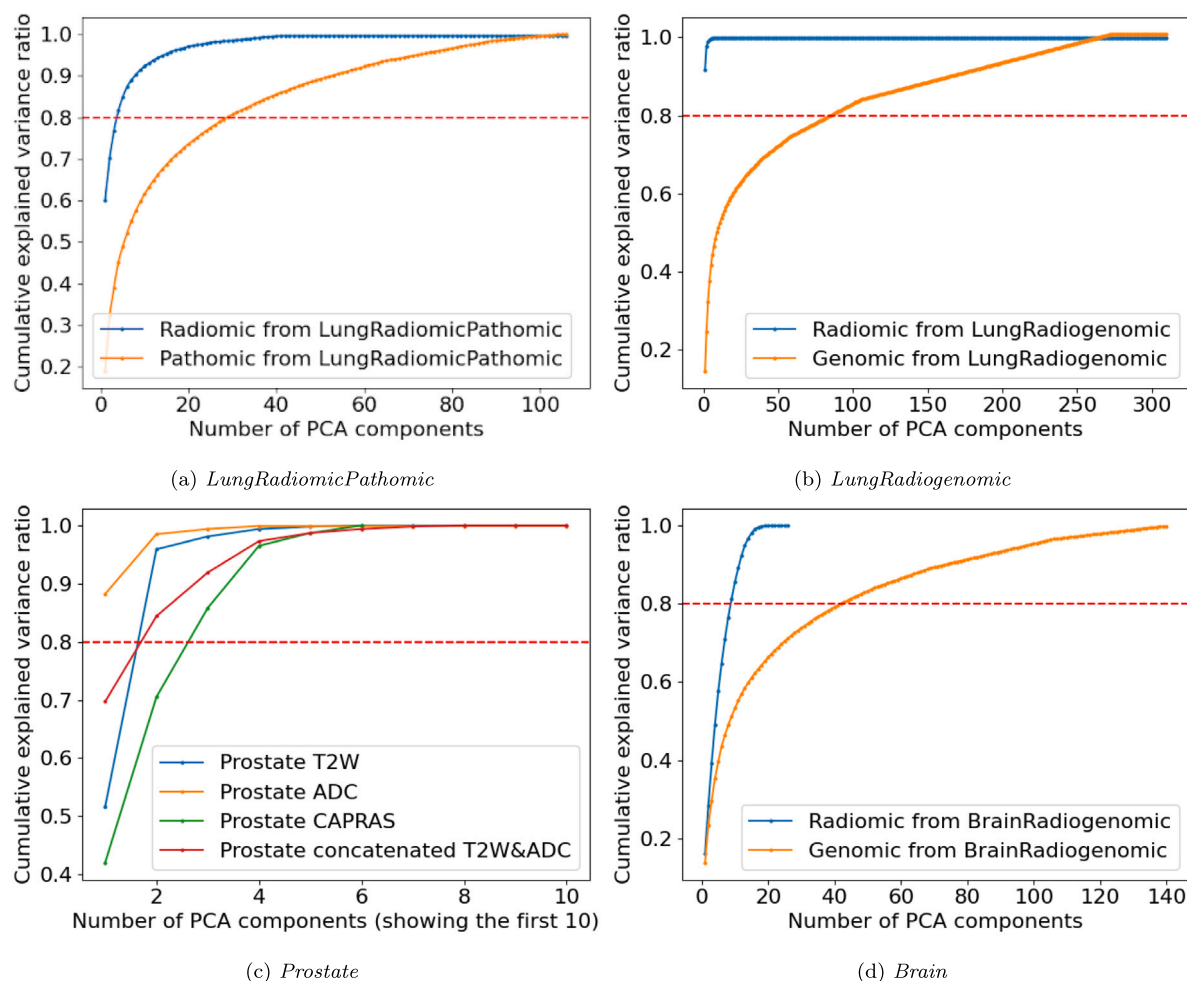


Fig. 7. The cumulative explained variance ratio of each modality in four cohorts. The number of components that resulted in at least 80% cumulative explained variance ratio was less than or equal to 5 for all modalities except for the pathomic data in *LungRadiomicPathomic*, the genomic data in *LungRadiogenomic*, the genomic data in *BrainRadiogenomic*, and the radiomic data in *LungRadiogenomic*, which had 30, 80, 45, and 9 components respectively. See Supplementary Tables 1, 3, 5, 7, 9, 11, and 13 for a complete view of the number of PCA components used in the study.

Liang et al. [5], we also quantified the degree of consistency between the PID values and model performance. We improved the PID metric calculation in two aspects. First, we found that the inconsistency might be partially related to how one chooses the preprocessing parameters when calculating the PID values. Although determining the optimal number of clusters for k-means clustering remains challenging, we discovered that minimizing information loss during PCA is beneficial. One approach to achieve this is by examining the cumulative explained variance ratio. Our second improvement to the PID calculation was that we added a 95% CI to each PID value, enabling one to estimate the uncertainty associated with each PID value. Presently, PID metrics are interpreted as relative values to see which interaction is most important within a dataset, but providing a method to compute PID in a bounded manner would improve the interpretability of the metrics. Further investigation to assess how PID metrics could assist in determining not only the generic category of whether early or late fusion should be used but also which specific fusion approach will be informative.

CRedit authorship contribution statement

Tengyue Zhang: Writing – review & editing, Visualization, Methodology, Formal analysis. **Ruiwen Ding:** Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Kha-Dinh Luong:** Writing – review & editing, Methodology. **William Hsu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Code availability

Code is available at: <https://github.com/zhtyolivia/pid-multimodal>. The PID metrics and deep learning models were implemented in Python; the linear models were implemented in R. The deep learning models were trained using NVIDIA Quadro RTX 8000 GPUs.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly to proofread the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jbi.2025.104833>.

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